HAROL MAXILAN

Data Science Undergraduate | Data Engineering & AI/ML Engineer

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PROFILE

Data Science undergraduate at Sabaragamuwa University of Sri Lanka specializing in Data Engineering and AI/ML Engineering. Hands-on experience designing Medallion lakehouse ETL pipelines (PySpark, AWS S3), training transformer NLP models (BERT, DistilBERT), and orchestrating multi-agent RAG systems with LangGraph. Seeking Data Engineering, AI/ML, or Data Science internship roles to build scalable data infrastructure and deploy robust intelligent systems.

EDUCATION

BSc (Hons) in Data Science — Sabaragamuwa University of Sri Lanka, Faculty of Computing *2024 – 2028 (Expected)*

Relevant Coursework: Database Systems, Data Warehousing, Data Science, Statistics, Computer Programming, Machine Learning, Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, PHP

Data Engineering & Big Data: ETL/ELT Pipelines, Medallion Architecture (Bronze–Silver–Gold), Apache Spark, PySpark, Batch Processing, Data Quality Auditing, Referential Integrity, Deduplication, Data Transformation, Parquet, Star/Snowflake Schema

Machine Learning & Deep Learning: Scikit-learn, PyTorch, Classification, Regression, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, SMOTE, XGBoost, AutoML, MLflow

Generative AI & LLMs: LangChain, LangGraph, Multi-Agent AI Systems, RAG (Retrieval-Augmented Generation), Prompt Engineering, Pydantic, Vector Search (FAISS, ChromaDB), Google Generative AI

Natural Language Processing (NLP): Hugging Face Transformers, BERT, DistilBERT, TF-IDF, Text Classification, Tokenization, Class Imbalance Handling

Computer Vision: OpenCV, MediaPipe, Hand Gesture Recognition, Facial Emotion Detection, CNNs, Real-Time Inference

Databases & Data Analysis: MySQL, PostgreSQL, MariaDB, Pandas, NumPy, Matplotlib, Plotly, Exploratory Data Analysis (EDA), Statistical Modeling

Cloud, DevOps & Tools: AWS (S3, EC2), FastAPI, Streamlit, Docker, Git, GitHub, Linux, Jupyter Notebook, VS Code, CI/CD, REST APIs, Pytest

PROJECTS & EXPERIENCE

CareerCompass AI — Autonomous Career Coaching System *Aug 2026*

Tech: *Python, LangGraph, LangChain, RAG, FAISS, Multi-Agent AI, FastAPI, Pydantic* | github.com/sankajithdjinasena/CareerCompass_AI

- Architected a Level 3 autonomous multi-agent career coaching ecosystem orchestrated via LangGraph state graphs with 5 collaborative AI agents (CV analyzer, gap assessor, interviewer, scraper, roadmap generator).
- Engineered a RAG pipeline utilizing FAISS vector embeddings and Tavily APIs to match candidate profiles against live job postings with Pydantic data validation.
- Designed deterministic JSON validation loops with self-correcting prompt mechanisms, achieving a 100% structured schema compliance rate across 50+ simulated mock interview sessions.

NexLab — Smart University Resource Allocation System *2025 – 2026*

Tech: *PHP, MySQL, JavaScript, HTML5/CSS3, Priority Scoring Engine, AI Intelligence Dashboard* | github.com/sankajithdjinasena/NEXLAB

- Led full-stack architecture as Team Leader (Team Predictra, CIPHER 2.0 Finalist) for an enterprise university lab resource allocation platform serving 500+ student/faculty requests.
- Engineered a multi-factor priority scoring engine with automated conflict resolution, round-robin slot allocation, and waitlist auto-promotion, eliminating 100% of double-booking conflicts across 4 user roles.
- Built an administrative Intelligence Dashboard delivering 7-day predictive demand forecasting, utilization heatmaps, and behavioral anomaly detection using dynamic threshold filters.

Syndicate DataStorm — Lakehouse Pipeline & Demand Prediction Model *2025*

Tech: *PySpark, Apache Spark, AWS S3, Medallion Architecture, Star Schema, Parquet, OpenStreetMap API* | github.com/Max-i26/Syndicate_DataStrom

- Designed and deployed a Medallion Lakehouse ETL pipeline using PySpark and AWS S3, transforming 2.4M+ transaction records into a Star Schema with Parquet storage across 20,000+ retail outlets.
- Built automated data quality gates enforcing referential integrity, deduplication, and schema validation, quarantining ~4.8% invalid records into Silver dead-letter tables.
- Extracted geospatial Point-of-Interest (POI) data via OpenStreetMap Overpass API, generating spatial density features that improved downstream retail demand forecasting RMSE by 18%.

NLP-Based Emotion Detection Pipeline *2025*

Tech: *Python, BERT, DistilBERT, MLflow, SMOTE, Scikit-learn, Hugging Face Transformers, PyTorch* | github.com/Max-i26/NLP-Emotion-Detection

- Engineered an end-to-end NLP classification pipeline benchmarking a TF-IDF baseline (0.76 F1) against fine-tuned BERT and DistilBERT transformer models.
- Tracked experiments and model versioning in MLflow; mitigated severe class imbalance across 20,000+ text samples using SMOTE, class-weighted loss, and label smoothing to boost minority-class Recall by 24%.
- Achieved a 0.88 Macro F1-score and 91.2% Weighted F1 on held-out test data, outperforming baseline models by 15.8% in multi-class emotional intent classification.

Friday — Automated Machine Learning Pipeline *2025*

Tech: *Python, Scikit-learn, XGBoost, Docker, FastAPI, Google Generative AI, Pandas* | github.com/Max-i26/Friday

- Architected an automated machine learning (AutoML) pipeline containerized with Docker and powered by a FastAPI backend for autonomous data ingestion and dynamic feature encoding.
- Built automated model benchmarking across XGBoost, LightGBM, and Random Forest models over 15+ benchmark datasets, cutting manual model selection time by 70%.
- Integrated Google Generative AI (Gemini API) to automatically synthesize diagnostic analytical reports and performance interpretations for stakeholders in <3 seconds.

Image-Based Emotion Detection *2025*

Tech: *Python, OpenCV, Deep Learning, CNN, Streamlit, h5py* | github.com/Max-i26/emotion_detection

- Engineered a real-time facial emotion recognition pipeline using OpenCV Haar Cascades for facial detection, bounding-box localization, and frame-level preprocessing.
- Trained and optimized a deep Convolutional Neural Network (CNN) across 35,000+ facial images (FER-2013), achieving 87.4% validation accuracy across 7 discrete emotional classes.
- Built an interactive Streamlit application supporting both batch image analysis and live webcam video streaming with <25ms real-time inference latency and dynamic confidence scores.

BEE — Hand Gesture-Controlled System Interface *2025*

Tech: *Python, MediaPipe, OpenCV, PyAutoGUI, Real-Time Computer Vision, HCI* | github.com/Max-i26/Bee

- Built a touchless human-computer interaction (HCI) interface using MediaPipe Hands and OpenCV, tracking 21 3D hand landmarks from live webcam streams at 30+ FPS with <20ms latency.
- Designed a geometric coordinate classifier achieving 96% recognition accuracy across 7 discrete multi-finger gestures (browser navigation, tab switching, media playback, sleep mode).
- Engineered an asynchronous action dispatch engine using PyAutoGUI and system-level APIs for cross-platform (Windows, macOS, Linux) desktop automation and shortcut execution.

F1 Super Intelligence — Data Analytics & Precomputation Pipeline *2025*

Tech: *Python, Pandas, Plotly, Pytest, Batch Processing, Data Quality Auditing* | github.com/Max-i26/F1-Super-Intelligence-System

- Constructed a high-performance batch ETL pipeline aggregating and precomputing 1.2M+ Formula 1 telemetry data points across 24 Grand Prix rounds for sub-second dashboard rendering.
- Developed a 56-test Pytest data auditing suite verifying sensor completeness, race strategies, and lap timing anomalies, achieving 100% telemetry integrity coverage.
- Built pace degradation and tire telemetry models rendering interactive Plotly dashboards, reducing analytical query response latency from 4.2s to <350ms.

AWARDS & ACHIEVEMENTS

- Nexora Ideathon — 3rd Place | Sabaragamuwa University of Sri Lanka (2025)
- CIPHER 2.0 Hackathon & Case Study — Finalist | Informatics Institute of Technology (IIT) (2025)
- J'PuraXtreme 3.0 Hackathon — Finalist | University of Sri Jayewardenepura (2025)
- CodeRally 7.0 Hackathon — Intermediate Tier Finalist | Informatics Institute of Technology (IIT) (2025)
- Code Arena Hackathon — Finalist | General Sir John Kotelawala Defence University (KDU) (2025)

LEADERSHIP & VOLUNTEER EXPERIENCE

- AWS Logistics and Operations Lead | Sabaragamuwa University of Sri Lanka (Oct 2026 – Present)
- Club Cricket Team Captain | VSC, Vavuniya (Oct 2025)
- Clash of Pawns Chess Competition Mentor | AUSDAV, Vavuniya (Jun 2025)
- Zone C University Ambassador — IEEE INSL | Sabaragamuwa University of Sri Lanka (Apr 2025)
- Pentathlon 2.0 Logistics Lead | AUSDAV, Vavuniya (Dec 2023)

REFERENCES

Professional and academic references available upon request.